

Swaham Pitambar Patra

Program:

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Python 3.8.5 Shell
emp-salary.py
1 import pandas as pd
2
3 df = pd.read_csv("employee_salary_dataset.csv")
4 print(df)
5
6 df = df.sort_values("Monthly_Salary")
7 print(df)
8
9 Q1 = df["Monthly_Salary"].quantile(0.25)
10 Q3 = df["Monthly_Salary"].quantile(0.75)
11
12 IQR = Q3 - Q1
13
14 outliers = df[
15     (df["Monthly_Salary"] < (Q1 - 1.5 * IQR)) |
16     (df["Monthly_Salary"] > (Q3 + 1.5 * IQR))
17 ]
18
19 print(outliers)
```

Output:

EmployeeID	Name	Department	Experience_Years	Education_Level	Age	Gender	City	Monthly_Salary
48	Employee_49	Finance	5	PhD	23	Female	Hyderabad	28420
13	Employee_14	Finance	6	Bachelor	27	Male	Delhi	28436
25	Employee_26	IT	14	Master	24	Male	Hyderabad	30600
16	Employee_17	Marketing	10	Master	41	Female	Hyderabad	30825
9	Employee_10	Operations	16	High School	28	Male	Mumbai	31891
8	Employee_9	IT	4	Bachelor	51	Male	Hyderabad	42420
38	Employee_39	HR	1	Master	24	Male	Mumbai	43589
47	Employee_48	Operations	1	High School	40	Male	Chennai	43711
17	Employee_18	Operations	12	PhD	51	Female	Chennai	44738
7	Employee_8	IT	17	PhD	24	Female	Bangalore	44828
18	Employee_19	HR	2	PhD	32	Male	Chennai	55746
44	Employee_45	Finance	8	Master	53	Male	Delhi	57635
46	Employee_47	Finance	8	Master	22	Female	Delhi	58828
41	Employee_42	HR	11	PhD	23	Male	Chennai	61212
23	Employee_24	HR	15	Master	40	Male	Bangalore	64790
28	Employee_29	Marketing	13	High School	48	Male	Bangalore	65704
6	Employee_6	Finance	3	High School	50	Male	Mumbai	69818
39	Employee_40	Finance	12	Bachelor	30	Female	Mumbai	69484
42	Employee_43	Marketing	19	High School	49	Female	Delhi	68525
2	Employee_3	IT	12	High School	51	Female	Hyderabad	69064
6	Employee_7	IT	14	PhD	57	Male	Mumbai	70525
30	Employee_31	IT	15	Bachelor	54	Female	Hyderabad	70714
43	Employee_44	Finance	17	Master	44	Male	Hyderabad	72302
12	Employee_13	Operations	2	Master	21	Female	Hyderabad	72318
20	Employee_21	Finance	4	Master	46	Female	Delhi	73404
19	Employee_20	Marketing	10	Bachelor	49	Male	Bangalore	74377
21	Employee_22	Operations	14	Master	54	Female	Delhi	79045
36	Employee_37	IT	7	High School	31	Male	Chennai	81491
31	Employee_32	Marketing	13	Master	45	Female	Hyderabad	81035
45	Employee_46	HR	3	PhD	54	Female	Hyderabad	88280
3	Employee_4	Operations	8	PhD	44	Male	Delhi	95091
1	Employee_2	Operations	7	Bachelor	25	Female	Bangalore	95271
35	Employee_36	Operations	1	Master	45	Female	Delhi	96744
14	Employee_15	Marketing	10	PhD	49	Female	Mumbai	99290
29	Employee_30	Marketing	18	High School	38	Male	Delhi	99543
15	Employee_16	Finance	4	High School	49	Female	Delhi	101213
11	Employee_12	Marketing	9	Master	49	Female	Hyderabad	104909
40	Employee_41	Marketing	8	Bachelor	23	Female	Bangalore	107989
27	Employee_28	Finance	16	Master	29	Female	Hyderabad	118179
26	Employee_27	IT	0	Bachelor	27	Female	Delhi	119660
49	Employee_50	Marketing	10	High School	47	Female	Bangalore	121752
22	Employee_23	Marketing	10	High School	22	Male	Bangalore	125888
32	Employee_33	Marketing	9	Master	36	Male	Bangalore	127045
34	Employee_35	IT	13	Master	53	Male	Bangalore	130983
4	Employee_5	Operations	15	Master	36	Female	Delhi	132458
24	Employee_25	HR	8	High School	34	Female	Bangalore	132458
33	Employee_34	HR	15	Bachelor	53	Female	Delhi	134616
10	Employee_11	Marketing	0	PhD	43	Male	Delhi	141301
37	Employee_38	Operations	9	Master	23	Male	Mumbai	149123
8	Employee_1	Marketing	18	Master	51	Female	Delhi	9111416
0	Employee_1	Marketing	15	Master	53	Female	Delhi	9111416

Program:

```
File Edit Selection View Go Help
Python 3.8.5

# Import pandas as pd
1 import pandas as pd
2
3 customers = pd.read_csv("customer_table.csv")
4 orders = pd.read_csv("orders_table.csv")
5
6 # Tables
7 print("Customer table")
8 print(customers)
9 print("Orders table")
10 print(orders)
11
12 # Left Join
13 l_join = pd.merge(customers, orders, on = "CustomerID", how = "left")
14 print("Left Join")
15 print(l_join)
16
17 # Inner Join
18 i_join = pd.merge(customers, orders, on = "CustomerID", how = "inner")
19 print("Inner Join")
20 print(i_join)
21
22 # Right Join
23 r_join = pd.merge(customers, orders, on = "CustomerID", how = "right")
24 print("Right Join")
25 print(r_join)
26
27 # Outer Join
28 o_join = pd.merge(customers, orders, on = "CustomerID", how = "outer")
29 print("Outer Join")
30 print(o_join)
```

Output:

```
PS C:\Users\Suham Patra\OneDrive\ITR Python\DS\Tasks 28-06\Cust-data> python cust-order.py

Customer Table
  CustomerID  OrderQuantity
0          1                2
1          2                3
2          3                4
3          4                5
4          5                6
5          6                7
6          7                8
7          8                9
8         10               10

Orders Table
  CustomerID  OrderAmount
0          1            12
1          2            23
2          4            45
3          5            56
4          7            67
5          8            78
6          9            89
7         10            90

Left Join
  CustomerID  OrderQuantity  OrderAmount
0          1                2           12.0
1          2                3           23.0
2          3                4           NaN
3          4                5           45.0
4          5                6           56.0
5          6                7           67.0
6          7                8           78.0
7          8                9           89.0
8         10               10           90.0

Inner Join
  CustomerID  OrderQuantity  OrderAmount
0          1                2           12
1          2                3           23
2          4                5           45
3          5                6           56
4          7                7           67
5          8                8           78
6          9                9           89
7         10               10           90

Right Join
  CustomerID  OrderQuantity  OrderAmount
0          1                2           12
1          2                3           23
2          4                5           45
3          5                6           56
4          7                7           67
5          8                8           78
6          9                9           89
7         10               10           90
```

```
Outer Join
CustomerID  OrderQuantity  OrderAmount
0           1           2           12.0
1           2           3           23.0
2           3           4           NaN
3           4           5           45.0
4           5           6           56.0
5           7           7           67.0
6           8           8           78.0
7           9           9           89.0
8          10          10           90.0
```

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 28-06\Cust-data>
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